

## Diagnosis information - Front-end electronics

### Wiper Voltage

#### **⚠ CAUTION**

Wiper motor start-up!

Danger of injury to hands and fingers when working on the wiper mechanism when the engine is running!

- ◆ Switch ignition off and/or disconnect the power supply for the wiper motor.

#### **NOTE**

Risk of damage to components!

Risk of damage to paintwork on the hood and wiper arms.

- ◆ The wiper arms may touch the hood if the hood is open and the wiper motor is running.
- ◆ Do not open the hood/engine compartment lid when the wiper arms are raised - risk of damage to paintwork and mechanism.

### Diagnostic conditions

The control unit must be active in order to test internal function sequences and diagnosable inputs/outputs.

In the event of a fault, check whether the fault is permanently >present<. To perform the check, erase the fault memory, meet the diagnostic conditions and read out the fault memory again.

- Ignition on

### Fault setting condition(s)

- Limit exceeded
- Below limit

### Fault effects

- ◆ **Wiper** No function

## Diagnosis information - Front-end electronics

### Wiper Voltage

#### **⚠ CAUTION**

Wiper motor start-up!

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Risk of damage to components!

Risk of damage to paintwork on the hood and wiper arms.

- ♦ The wiper arms may touch the hood if the hood is open and the wiper motor is running.
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The control unit must be active in order to test internal function sequences and diagnosable inputs/outputs.

In the event of a fault, check whether the fault is permanently >present<. To perform the check, erase the fault memory, meet the diagnostic conditions and read out the fault memory again.

- Ignition on

### Fault setting condition(s)

- Limit exceeded
- Below limit

### Fault effects

- ♦ **Wiper** No function

## Diagnosis information - Front-end electronics

### Wiper overload

#### **⚠ CAUTION**

Wiper motor start-up!

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- ◆ Switch ignition off and/or disconnect the power supply for the wiper motor.

#### **NOTE**

Risk of damage to components!

Risk of damage to paintwork on the hood and wiper arms.

- ◆ The wiper arms may touch the hood if the hood is open and the wiper motor is running.
- ◆ Do not open the hood/engine compartment lid when the wiper arms are raised - risk of damage to paintwork and mechanism.

### Diagnostic conditions

The control unit must be active in order to test internal function sequences and diagnosable inputs/outputs.

In the event of a fault, check whether the fault is permanently >present<. To perform the check, erase the fault memory, meet the diagnostic conditions and read out the fault memory again.

- Ignition on

### Fault setting condition(s)

- Overload protection: Permanently above limit value (> 4 seconds)

### Possible fault causes

- ◆ **Wiper motor** blocked / Faulty
- ◆ **Wiper linkage** blocked / mechanically sluggish
- ◆ **Wiper arm** blocked / mechanically sluggish

## Diagnosis information - Front-end electronics

### Wiper no communication

#### ⚠ CAUTION

Wiper motor can come on!

Danger of injury to hands and fingers when working on the wiper mechanism when the engine is running!

- ◆ Switch ignition off and/or disconnect the power supply for the wiper motor.

#### NOTE

Risk of damage to parts!

Risk of damage to paintwork on the hood and wiper arms.

- ◆ The wiper arms can touch the hood if the hood is open and the wiper motor is running.
- ◆ Do not open the hood/engine compartment lid when the wiper arms are raised - risk of damage to paintwork and mechanism.

### Diagnostic conditions

The control unit must be active in order to test internal function sequences and diagnosable inputs/outputs.

In the event of a fault, check whether the fault is permanently >present<. To perform the check, erase the fault memory, meet the diagnostic conditions and read out the fault memory again.

- Ignition on

### Possible fault causes

- ◆ No power supply for the **Wiper** or only limited power supply available.
- ◆ Open circuit in LIN line between the **Front-end electronics** control unit and the **Wiper**
- ◆ **Wiper** faulty

### Fault setting condition(s)

- No communication with the **Wiper** for  $t > 4 \text{ s}$

### Fault effects

- **Wiper**: Function not available or available only to a limited extent



## Diagnosis information - Front-end electronics

### Wiper Faulty

#### ⚠ CAUTION

Wiper motor can come on!

Danger of injury to hands and fingers when working on the wiper mechanism when the engine is running!

- ◆ Switch ignition off and/or disconnect the power supply for the wiper motor.

#### NOTE

Risk of damage to parts!

Risk of damage to paintwork on the hood and wiper arms.

- ◆ The wiper arms can touch the hood if the hood is open and the wiper motor is running.
- ◆ Do not open the hood/engine compartment lid when the wiper arms are raised - risk of damage to paintwork and mechanism.

### Diagnostic conditions

The control unit must be active in order to test internal function sequences and diagnosable inputs/outputs.

In the event of a fault, check whether the fault is permanently >present<. To perform the check, erase the fault memory, meet the diagnostic conditions and read out the fault memory again.

- Ignition on

### Possible fault causes

- ◆ Wiper Faulty

### Fault setting condition(s)

- Wiper reports fault for  $t > 4 \text{ s}$

### Fault effects

**Wiper** No function

## Diagnosis information - Front-end electronics

### Window washer system pump Faulty

#### Diagnostic conditions

The control unit must be active in order to test internal function sequences and diagnosable inputs/outputs.

In the event of a fault, check whether the fault is permanently >present<. To perform the check, erase the fault memory, meet the diagnostic conditions and read out the fault memory again.

#### Fault setting condition(s)

- Ignition on

#### Fault effects

**Window washer system pump** No function

## Diagnosis information - Front-end electronics

### Rain sensor no communication

#### Diagnostic conditions

The control unit must be active in order to test internal function sequences and diagnosable inputs/outputs.

In the event of a fault, check whether the fault is permanently >present<. To perform the check, erase the fault memory, meet the diagnostic conditions and read out the fault memory again.

- Ignition on
- **Rain sensor** coded

#### Possible fault causes

- ♦ No power supply for the **Rain sensor** or only limited power supply available.
- ♦ Open circuit in LIN line between the **Front-end electronics** control unit and the **Rain sensor**
- ♦ **Rain sensor** faulty

#### Fault setting condition(s)

- No communication with the **Rain sensor** for  $t > 4$  s

#### Fault effects

- **Rain sensor** No function

## Diagnosis information - Front-end electronics

**Electronic steering column lock (ELV), Supply voltage (Terminal 30) Below limit**

### Diagnostic conditions

The control unit must be active in order to test internal function sequences and diagnosable inputs/outputs.

In the event of a fault, check whether the fault is permanently >present<. To perform the check, erase the fault memory, meet the diagnostic conditions and read out the fault memory again.

- Ignition on

### Possible fault causes

- ♦ Open circuit/short circuit to ground
- ♦ Fuse Faulty

### Fault setting condition(s)

- **Supply voltage** Below limit

### Fault effects

- **Electronic steering column lock (ELV)** Function not available or available only to a limited extent

## Diagnosis information - Front-end electronics

### Electronic steering column lock (ELV) Terminal 15

#### Diagnostic conditions

The control unit must be active in order to test internal function sequences and diagnosable inputs/outputs.

In the event of a fault, check whether the fault is permanently >present<. To perform the check, erase the fault memory, meet the diagnostic conditions and read out the fault memory again.

#### Fault setting condition(s)

- Ignition on
  - Below limit
- The measured value does **not** correspond to setpoint specification

#### Fault effects

**Electronic steering column lock (ELV)** not locked

## Diagnosis information - Front-end electronics

### Electronic steering column lock (ELV) Engine Voltage

#### Diagnostic conditions

The control unit must be active in order to test internal function sequences and diagnosable inputs/outputs.

In the event of a fault, check whether the fault is permanently >present<. To perform the check, erase the fault memory, meet the diagnostic conditions and read out the fault memory again.

- Ignition on

#### Possible fault causes

- ♦ Short circuit to B+/ground or open circuit in the line (supply line) between **Front-end electronics** <-> Engine **Electronic steering column lock (ELV)**

#### Fault effects

**Electronic steering column lock (ELV) Engine** No function

## Diagnosis information - Front-end electronics

### Electronic steering column lock (ELV) Engine Position/s implausible

#### Diagnostic conditions

The control unit must be active in order to test internal function sequences and diagnosable inputs/outputs.

In the event of a fault, check whether the fault is permanently >present<. To perform the check, erase the fault memory, meet the diagnostic conditions and read out the fault memory again.

#### Fault setting condition(s)

- Ignition on

#### Fault effects

Electronic steering column lock (ELV) Engine No function

## Diagnosis information - Front-end electronics

### Electronic steering column lock (ELV) Engine Voltage

#### Diagnostic conditions

The control unit must be active in order to test internal function sequences and diagnosable inputs/outputs.

In the event of a fault, check whether the fault is permanently >present<. To perform the check, erase the fault memory, meet the diagnostic conditions and read out the fault memory again.

- Ignition on

#### Possible fault causes

- ♦ Short circuit to ground in the Power supply

#### Fault effects

**Electronic steering column lock (ELV) Engine** No function

## Diagnosis information - Front-end electronics

### Locking motor

### Diagnostic conditions

The control unit must be active in order to test internal function sequences and diagnosable inputs/outputs.

In the event of a fault, check whether the fault is permanently >present<. To perform the check, erase the fault memory, meet the diagnostic conditions and read out the fault memory again.

### Fault setting condition(s)

- Ignition on
- Communication **Electronic steering column lock (ELV) with Front-end electronics**

### Fault effects

**Electronic steering column lock (ELV)** No function

## Diagnosis information - Front-end electronics

**Electronic steering column lock (ELV) Switch implausible**

### Diagnostic conditions

The control unit must be active in order to test internal function sequences and diagnosable inputs/outputs.

In the event of a fault, check whether the fault is permanently >present<. To perform the check, erase the fault memory, meet the diagnostic conditions and read out the fault memory again.

### Fault setting condition(s)

- Ignition on
- **Electronic steering column lock (ELV) Switch Position/s implausible**

### Fault effects

**Electronic steering column lock (ELV) Engine No function**

## Diagnosis information - Front-end electronics

**Electronic steering column lock (ELV) Switch implausible**

### Diagnostic conditions

The control unit must be active in order to test internal function sequences and diagnosable inputs/outputs.

In the event of a fault, check whether the fault is permanently >present<. To perform the check, erase the fault memory, meet the diagnostic conditions and read out the fault memory again.

### Fault setting condition(s)

- Ignition on
- **Electronic steering column lock (ELV) Switch Position/s implausible**

### Fault effects

**Electronic steering column lock (ELV) Engine No function**

## Diagnosis information - Front-end electronics

### Engine Electronic steering column lock (ELV) Faulty

#### Diagnostic conditions

The control unit must be active in order to test internal function sequences and diagnosable inputs/outputs.

In the event of a fault, check whether the fault is permanently >present<. To perform the check, erase the fault memory, meet the diagnostic conditions and read out the fault memory again.

- Ignition on

#### Possible fault causes

- ♦ Engine **Electronic steering column lock (ELV)** Faulty

#### Fault effects

Engine **Electronic steering column lock (ELV)** No function

## Diagnosis information - Front-end electronics

### Electronic steering column lock (ELV) Engine Short circuit Terminal 30

#### Diagnostic conditions

The control unit must be active in order to test internal function sequences and diagnosable inputs/outputs.

In the event of a fault, check whether the fault is permanently >present<. To perform the check, erase the fault memory, meet the diagnostic conditions and read out the fault memory again.

#### Fault setting condition(s)

- Ignition on
- **Electronic steering column lock (ELV)** Engine Power or ground supply faulty

#### Fault effects

**Electronic steering column lock (ELV)** Engine No function

## Diagnosis information - Front-end electronics

### Electronic steering column lock (ELV) Engine Short circuit Terminal 30

#### Diagnostic conditions

The control unit must be active in order to test internal function sequences and diagnosable inputs/outputs.

In the event of a fault, check whether the fault is permanently >present<. To perform the check, erase the fault memory, meet the diagnostic conditions and read out the fault memory again.

#### Fault setting condition(s)

- Ignition on
- **Electronic steering column lock (ELV)** Engine Power or ground supply faulty

#### Fault effects

**Electronic steering column lock (ELV)** Engine No function

## Diagnosis information - Front-end electronics

### Electronic steering column lock (ELV) Faulty

#### Diagnostic conditions

The control unit must be active in order to test internal function sequences and diagnosable inputs/outputs.

In the event of a fault, check whether the fault is permanently >present<. To perform the check, erase the fault memory, meet the diagnostic conditions and read out the fault memory again.

- Ignition on

#### Possible fault causes

- ♦ Electronic steering column lock (ELV) Faulty

#### Fault effects

Electronic steering column lock (ELV) No function

## Diagnosis information - Front-end electronics

### Electronic steering column lock (ELV) Motor current Limit exceeded

#### Diagnostic conditions

The control unit must be active in order to test internal function sequences and diagnosable inputs/outputs.

In the event of a fault, check whether the fault is permanently >present<. To perform the check, erase the fault memory, meet the diagnostic conditions and read out the fault memory again.

#### Fault setting condition(s)

- Current , - The measured value does **not** correspond to setpoint specification

#### Fault effects

**Electronic steering column lock (ELV)** Engine No function

## Diagnosis information - Front-end electronics

### Electronic steering column lock (ELV) Motor current Limit exceeded

#### Diagnostic conditions

The control unit must be active in order to test internal function sequences and diagnosable inputs/outputs.

In the event of a fault, check whether the fault is permanently >present<. To perform the check, erase the fault memory, meet the diagnostic conditions and read out the fault memory again.

#### Fault setting condition(s)

- Current , - The measured value does **not** correspond to setpoint specification

#### Fault effects

**Electronic steering column lock (ELV)** Engine No function

## Diagnosis information - Front-end electronics

### Electronic steering column lock (ELV) Faulty

#### Diagnostic conditions

The control unit must be active in order to test internal function sequences and diagnosable inputs/outputs.

In the event of a fault, check whether the fault is permanently >present<. To perform the check, erase the fault memory, meet the diagnostic conditions and read out the fault memory again.

- Ignition on

#### Possible fault causes

- ♦ Electronic steering column lock (ELV) Faulty

#### Fault effects

Electronic steering column lock (ELV) No function

## Diagnosis information - Front-end electronics

### Electronic steering column lock (ELV) Faulty

#### Diagnostic conditions

The control unit must be active in order to test internal function sequences and diagnosable inputs/outputs.

In the event of a fault, check whether the fault is permanently >present<. To perform the check, erase the fault memory, meet the diagnostic conditions and read out the fault memory again.

- Ignition on

#### Possible fault causes

- ♦ Electronic steering column lock (ELV) Faulty

#### Fault effects

Electronic steering column lock (ELV) No function

## Diagnosis information - Front-end electronics

**Electronic steering column lock (ELV) Enabling line, Electrical fault**

### Diagnostic conditions

The control unit must be active in order to test internal function sequences and diagnosable inputs/outputs.

In the event of a fault, check whether the fault is permanently >present<. To perform the check, erase the fault memory, meet the diagnostic conditions and read out the fault memory again.

- Ignition on

### Possible fault causes

- ♦ Short circuit to B+
- ♦ Short circuit to ground

### Fault setting condition(s)

- **Electronic steering column lock (ELV) Enabling line** Faulty

### Fault effects

**Electronic steering column lock (ELV)** No function

## Diagnosis information - Front-end electronics

**Electronic steering column lock (ELV) Enabling line, Electrical fault**

### Diagnostic conditions

The control unit must be active in order to test internal function sequences and diagnosable inputs/outputs.

In the event of a fault, check whether the fault is permanently >present<. To perform the check, erase the fault memory, meet the diagnostic conditions and read out the fault memory again.

- Ignition on

### Possible fault causes

- ♦ Short circuit to B+
- ♦ Short circuit to ground

### Fault setting condition(s)

- **Electronic steering column lock (ELV) Enabling line** Faulty

### Fault effects

**Electronic steering column lock (ELV)** No function

## Diagnosis information - Front-end electronics

### Electronic steering column lock (ELV) Motor current Limit exceeded

#### Diagnostic conditions

The control unit must be active in order to test internal function sequences and diagnosable inputs/outputs.

In the event of a fault, check whether the fault is permanently >present<. To perform the check, erase the fault memory, meet the diagnostic conditions and read out the fault memory again.

#### Fault setting condition(s)

- Current , - The measured value does **not** correspond to setpoint specification

#### Fault effects

**Electronic steering column lock (ELV)** Engine No function

## Diagnosis information - Front-end electronics

### Relay stuck

#### Diagnostic conditions

The control unit must be active in order to test internal function sequences and diagnosable inputs/outputs.

In the event of a fault, check whether the fault is permanently >present<. To perform the check, erase the fault memory, meet the diagnostic conditions and read out the fault memory again.

#### Fault setting condition(s)

- Ignition off
- Feedback activated but Relay **not** activated

#### Fault effects

Engine start **not** possible

## Diagnosis information - Front-end electronics

### Relay stuck

#### Diagnostic conditions

The control unit must be active in order to test internal function sequences and diagnosable inputs/outputs.

In the event of a fault, check whether the fault is permanently >present<. To perform the check, erase the fault memory, meet the diagnostic conditions and read out the fault memory again.

#### Fault setting condition(s)

- Ignition off
- Feedback activated but Relay **not** activated

#### Fault effects

Engine start **not** possible

## Diagnosis information - Front-end electronics

### Relay stuck

#### Diagnostic conditions

The control unit must be active in order to test internal function sequences and diagnosable inputs/outputs.

In the event of a fault, check whether the fault is permanently >present<. To perform the check, erase the fault memory, meet the diagnostic conditions and read out the fault memory again.

#### Fault setting condition(s)

- Ignition off
- Feedback activated but Relay **not** activated

#### Fault effects

Engine start **not** possible

## Diagnosis information - Front-end electronics

### Relay stuck

#### Diagnostic conditions

The control unit must be active in order to test internal function sequences and diagnosable inputs/outputs.

In the event of a fault, check whether the fault is permanently >present<. To perform the check, erase the fault memory, meet the diagnostic conditions and read out the fault memory again.

#### Fault setting condition(s)

- Ignition off
- Feedback activated but Relay **not** activated

#### Fault effects

Engine start **not** possible

## Diagnosis information - Front-end electronics

### Relay stuck

#### Diagnostic conditions

The control unit must be active in order to test internal function sequences and diagnosable inputs/outputs.

In the event of a fault, check whether the fault is permanently >present<. To perform the check, erase the fault memory, meet the diagnostic conditions and read out the fault memory again.

#### Fault setting condition(s)

- Ignition off
- Feedback activated but Relay **not** activated

#### Fault effects

Engine start **not** possible

## Diagnosis information - Front-end electronics

### Relay stuck

#### Diagnostic conditions

The control unit must be active in order to test internal function sequences and diagnosable inputs/outputs.

In the event of a fault, check whether the fault is permanently >present<. To perform the check, erase the fault memory, meet the diagnostic conditions and read out the fault memory again.

#### Fault setting condition(s)

- Ignition off
- Feedback activated but Relay **not** activated

#### Fault effects

Engine start **not** possible

## Diagnosis information - Front-end electronics

### Output Terminal 15 SV implausible

#### Diagnostic conditions

The control unit must be active in order to test internal function sequences and diagnosable inputs/outputs.

In the event of a fault, check whether the fault is permanently >present<. To perform the check, erase the fault memory, meet the diagnostic conditions and read out the fault memory again.

#### Fault setting condition(s)

- Ignition on
- Current **Control module** Output Too high

#### Fault effects

Engine start **not** possible

## Diagnosis information - Front-end electronics

### Keys Position/s implausible

#### Diagnostic conditions

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In the event of a fault, check whether the fault is permanently >present<. To perform the check, erase the fault memory, meet the diagnostic conditions and read out the fault memory again.

#### Fault setting condition(s)

- Keys in the **Ignition lock**
- Switch **Ignition lock** internal , Position implausible

#### Fault effects

Engine start **not** possible

## Diagnosis information - Front-end electronics

**Electronic ignition starting switch Short circuit to B+/ground or open circuit in the line**

### Diagnostic conditions

The control unit must be active in order to test internal function sequences and diagnosable inputs/outputs.

In the event of a fault, check whether the fault is permanently >present<. To perform the check, erase the fault memory, meet the diagnostic conditions and read out the fault memory again.

- Keys in the **Ignition lock**

### Possible fault causes

- ♦ Switch in the **Ignition lock** faulty
- ♦ Short circuit to B+/ground or open circuit in the line

### Fault effects

Engine start **not** possible

## Diagnosis information - Front-end electronics

**Electronic ignition starting switch Short circuit to B+/ground or open circuit in the line**

### Diagnostic conditions

The control unit must be active in order to test internal function sequences and diagnosable inputs/outputs.

In the event of a fault, check whether the fault is permanently >present<. To perform the check, erase the fault memory, meet the diagnostic conditions and read out the fault memory again.

- Keys in the **Ignition lock**

### Possible fault causes

- ♦ Switch in the **Ignition lock** faulty
- ♦ Short circuit to B+/ground or open circuit in the line

### Fault effects

Engine start **not** possible

## Diagnosis information - Front-end electronics

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The control unit must be active in order to test internal function sequences and diagnosable inputs/outputs.

In the event of a fault, check whether the fault is permanently >present<. To perform the check, erase the fault memory, meet the diagnostic conditions and read out the fault memory again.

- Keys in the **Ignition lock**

### Possible fault causes

- ♦ Switch in the **Ignition lock** faulty
- ♦ Short circuit to B+/ground or open circuit in the line

### Fault effects

Engine start **not** possible

## Diagnosis information - Front-end electronics

**Electronic ignition starting switch Short circuit to B+/ground or open circuit in the line**

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The control unit must be active in order to test internal function sequences and diagnosable inputs/outputs.

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- Keys in the **Ignition lock**

### Possible fault causes

- ♦ Switch in the **Ignition lock** faulty
- ♦ Short circuit to B+/ground or open circuit in the line

### Fault effects

Engine start **not** possible

## Diagnosis information - Front-end electronics

**Electronic ignition starting switch Short circuit to B+/ground or open circuit in the line**

### Diagnostic conditions

The control unit must be active in order to test internal function sequences and diagnosable inputs/outputs.

In the event of a fault, check whether the fault is permanently >present<. To perform the check, erase the fault memory, meet the diagnostic conditions and read out the fault memory again.

- Keys in the **Ignition lock**

### Possible fault causes

- ♦ Switch in the **Ignition lock** faulty
- ♦ Short circuit to B+/ground or open circuit in the line

### Fault effects

Engine start **not** possible